Pandas Interview Questions with Dataset and Solutions

Sample Dataset:

```
import pandas as pd

data = {
   "EmpID": [1, 2, 3, 4, 5, 6, 7],
   "Name": ["Anand", "Riya", "Karan", "Sneha", "Amit", "Neha", "Raj"],
   "Dept": ["IT", "HR", "IT", "Finance", "HR", "IT", "Finance"],
   "Salary": [70000, 50000, 80000, 60000, 52000, 90000, 65000],
   "Experience": [5, 3, 7, 4, 2, 8, 6]
}

df = pd.DataFrame(data)
```

Top 25 Pandas Interview Questions with Solutions:

1. Show first 5 rows

```
df.head()
```

2. Get DataFrame shape

```
df.shape
```

3. Select only Name and Salary

df[["Name", "Salary"]]

4. Employees with Salary > 60000

df[df["Salary"] > 60000]

5. Employees in IT department

df[df["Dept"] == "IT"]

6. Sort by Salary descending

df.sort_values("Salary", ascending=False)

7. Average salary per department

df.groupby("Dept")["Salary"].mean()

8. Total salary per department

df.groupby("Dept")["Salary"].sum()

9. Count employees per department

df["Dept"].value_counts()

10. Add bonus column = 10% of salary

df["Bonus"] = df["Salary"] * 0.10

11. Highest salary employee

df.loc[df["Salary"].idxmax()]

12. Second highest salary

```
df["Salary"].nlargest(2).iloc[-1]
```

13. Top 2 salaries

```
df.nlargest(2, "Salary")
```

14. Top salary per department

```
df.loc[df.groupby("Dept")["Salary"].idxmax()]
```

15. Running total of salary by department

```
df["RunTotal"] = df.groupby("Dept")["Salary"].cumsum()
```

16. Rank salary within department

```
df["Rank"] = df.groupby("Dept")["Salary"].rank(method="dense", ascending=False)
```

17. Previous salary in same department (LAG)

```
df["PrevSalary"] = df.groupby("Dept")["Salary"].shift(1)
```

18. Next salary (LEAD)

```
df["NextSalary"] = df.groupby("Dept")["Salary"].shift(-1)
```

19. Employees with experience > department average

```
dept_avg = df.groupby("Dept")["Experience"].transform("mean")

df[df["Experience"] > dept_avg]
```

20. Multiple aggregations

```
df.groupby("Dept").agg({
    "Salary": ["min", "max", "mean"],
    "EmpID": "count"
})
```

21. Create salary grade column

```
df["Grade"] = df["Salary"].apply(
    lambda x: "High" if x > 70000 else "Medium" if x > 55000 else "Low"
)
```

22. Pivot table of average salary per department

```
pd.pivot_table(df, values="Salary", index="Dept", aggfunc="mean")
```

23. Rename column Dept → Department

```
df.rename(columns={"Dept": "Department"}, inplace=True)
```

24. Drop Experience column

```
df.drop(columns=["Experience"], inplace=True)
```

25. Export to CSV

```
df.to_csv("employees.csv", index=False)
```